

## **Supervised credit-risk prediction**

### **1. Abstract**

This report captures the supervised learning part of the analysis for the loan dataset so we can predict whether the loan ends in a good outcome or a default. To keep the problem well defined we restricted the target variable to two final statuses: Fully Paid and Charged Off. The analysis uses borrower and loan characteristics observed at origination, including loan size, interest rate, installment, annual income, debt-to-income ratio, revolving-credit usage, credit-history length, categorical descriptors such as grade, employment length, home ownership, verification status, and loan purpose. After cleaning and preparing the data two classification methods were chosen: Linear Discriminant Analysis (LDA) as a simple benchmark and Random Forest as the main nonlinear model. Since the dataset is imbalanced, with many more good loans than bad loans, special attention was given to evaluation beyond raw accuracy. The results show that the benchmark models with standard classification rules were too conservative in identifying bad loans, while a tuned Random Forest achieved a much better balance between detecting charged-off loans and limiting false alarms. Overall, the supervised analysis shows that loan default risk can be meaningfully predicted from origination-time information, but that model flexibility and careful threshold selection are important for obtaining practically useful results.

### **2. Statement of the problem / goal of the analysis and description of the dataset**

The goal of the supervised part is to see whether we can predict future loan outcomes based on the information available at the moment of the origination of the loan based on the individuals characteristics. Unlike the unsupervised part where we focused on finding borrower segments, here our purpose is more straightforward, which is to test whether borrower and loan characteristics at origination contain enough useful information to distinguish between loans that end well and loans that end badly. In simple terms, the goal is to find out whether it is possible to identify risky loans early before their final outcome is known.

The analysis is based on the Loan Data 2007–2014 dataset. For the supervised setting, the target variable was defined in the cleanest possible way by keeping only loans with two clear final outcomes - Fully Paid and Charged Off. These were treated as the two classes of the prediction problem, where fully paid ones represent loans that were repaid and charged off represent credit loss for the lender. Other loan statuses were excluded because they are less suitable for building a stable and interpretable classification model leading to vague results. The predictors used in the analysis describe both the loan itself and the borrower's financial position at origination. Including variables such as loan amount, loan term, interest rate, installment, annual income, debt-to-income ratio, delinquencies in the previous two years, recent credit inquiries, number of open accounts, public records, revolving balance, revolving utilization, total number of accounts, employment length, home ownership, verification status, loan purpose, and initial listing status. Next we created a measure of credit length timeline from the issue date and the borrower's earliest recorded credit line. Loan grade was

also retained because it is assigned at origination and summarizes part of the lender's initial assessment of credit quality.

Before modelling, the data was cleaned and transformed so that the prediction task would be based on a consistent and usable set of variables. Numeric values stored as text were converted into proper numeric form, percentage signs and commas were removed where necessary, loan term was rewritten as a numeric number of months, and employment length was standardized into consistent categories. Also some monetary variables were right-skewed, loan amount, installment, annual income, and revolving balance were log-transformed to reduce the influence of extreme values. At the end the final supervised dataset contained 226,540 loans. The modelling stage compares Linear Discriminant Analysis (LDA) as a simple benchmark and Random Forest as the main nonlinear model, because the dataset contains more loans with positive outcomes than negative ones, it cannot be judged only by the accuracy of it. So the analysis pays particular attention to how well the models identify bad loans, using measures such as precision, recall, and the F1 score for the default loans.

### **3. Findings / key points**

#### **1. Origination-time borrower and loan information contains real predictive value.**

The results reveal future loan performance is not random based on the information available at origination. These variables: loan sizes, installment burden, interest rate, income, debt-to-income ratio, revolving-debt use, credit-history depth contain enough signal to help distinguish between loans that are eventually repaid and loans that end in charge-off. The dataset supports a meaningful supervised prediction task rather than a purely descriptive exercise.

#### **2. Credit risk in this portfolio reflects a combination of loan burden, pricing, and existing credit use, rather than just borrower income or lender grade.**

The Random Forest importance ranking shows that the strongest predictors were loan amount, installment, interest rate, grade, income, revolving balance, loan term, revolving utilization, and debt-to-income ratio. Charged-off loans are based on the interaction between how much is borrowed, how expensive the loan is, and how stretched the borrower already appears to be and not just with lower income or weaker credit labels. This finding makes the result more informative since it avoids the safe versus risky interpretation, and reveals a more multidimensional financial meaning.

#### **3. In this imbalanced credit dataset, default model settings were too conservative to identify charged-off loans effectively.**

Both LDA and the Random Forest with the standard 0.50 threshold achieved relatively high overall accuracy but it detected only a small fraction of charged-off loans. After lowering the Random Forest threshold to 0.25, bad-loan recall increased, although at the cost of more false positives. This shows that for this dataset, model

usefulness depended not only on the algorithm, but also on choosing a decision rule that matched our goal of risk screening.

#### 4. Analysis with commentary

LDA was used as a simple benchmark, while Random Forest was used as the main nonlinear classifier. The comparison was testing whether a simple linear rule was sufficient or whether a more flexible model was needed. The first important result is that overall accuracy is not enough to judge model quality in this dataset. Because fully paid loans are much more common than charged-off loans, a model can achieve a high accuracy while still performing poorly in detecting charged off loans as we uncovered.

**Table 1. Comparison of supervised models**

| Model                             | Accuracy | Precision<br>(bad) | Recall<br>(bad) | Specificity | F1<br>(bad) |
|-----------------------------------|----------|--------------------|-----------------|-------------|-------------|
| LDA                               | 0.8126   | 0.4897             | 0.0868          | 0.9792      | 0.1474      |
| Random Forest (0.50<br>threshold) | 0.8142   | 0.5308             | 0.0428          | 0.9913      | 0.0792      |
| Random Forest (0.25<br>threshold) | 0.7120   | 0.3303             | 0.5280          | 0.7542      | 0.4064      |

As shown in **Table 1**, both LDA and the untuned Random Forest achieved relatively high accuracy, but this came mainly from classifying the majority class correctly. In practical terms, both models were too conservative and missed most charged-off loans. LDA detected only about **8.7%** of the bad loans, while the Random Forest with the default 0.50 threshold detected only about **4.3%**. For a credit-risk application, this is too weak to be useful. For this reason, the Random Forest threshold was adjusted instead of being left at its default value. Once the threshold was lowered to **0.25**, the model became much more effective at identifying problematic loans. Recall for the bad class increased to about **52.8%**, and the F1 score improved substantially. The cost of this improvement was a lower overall accuracy and more false positives, meaning that more fully paid loans were also flagged as risky. This trade-off is important. The final model is therefore better interpreted as a screening and prioritization tool than as a fully automatic approval rule. We could tune the threshold more to increase the recall for the bad class, where for example 0.10 threshold would yield 92%, however it also increases the number of false positives by a lot flagging the paid off loans as faulty ones which would lower the credibility of the analysis. The next question is whether the model is relying on economically meaningful predictors. The variable-importance ranking suggests that it is.

**Table 2. Most important predictors in the final Random Forest model**

| Rank | Predictor       | What it captures in simple terms           |
|------|-----------------|--------------------------------------------|
| 1    | log_loan_amnt   | overall loan size                          |
| 2    | log_installment | monthly repayment burden                   |
| 3    | int_rate        | loan pricing / perceived borrower risk     |
| 4    | grade           | lender's initial credit-quality assessment |
| 5    | log_annual_inc  | borrower earning capacity                  |
| 6    | log_revol_bal   | existing revolving debt burden             |
| 7    | term_num        | loan maturity                              |
| 8    | revol_util      | intensity of revolving-debt use            |
| 9    | total_acc       | depth of credit history                    |
| 10   | dti             | debt burden relative to income             |

As shown in **Table 2**, the model is driven by a combination of borrowing scale, repayment burden, pricing, income, existing debt pressure, and credit-history characteristics. That is a sensible result, which suggests that future loan performance is explained by the interaction of several financial features observed at origination. Overall, the supervised results show that the dataset does contain meaningful predictive information, but a simple benchmark model is not enough. The best-performing specification in the project was the Random Forest with a tuned threshold of 0.25, because it provided the most useful balance between identifying charged-off loans and limiting false alarms. From a practical point of view, that makes it the strongest model in the project for risk screening.

## **6. Conclusion**

The results show that the dataset does contain meaningful predictive information, although that model usefulness depends on how the performance of the model is evaluated. A simple benchmark model such as LDA achieved relatively high overall accuracy, yet it failed to identify most charged-off loans, which let us know that the raw accuracy wasn't an adequate measure of success in an imbalanced credit-risk setting. The Random Forest model provided a stronger basis for prediction. The default threshold missed most of the needed values and after tuning the classification threshold on the validation set, the preferred specification model was with a threshold of 0.25. This model produced the best balance between detecting charged-off loans and limiting false alarms, achieving a much higher recall for the bad class than the benchmark models. So we chose this model as the main basis for the research. The variable importance results showed that the model relied on financially meaningful predictors

such as loan size, installment burden, interest rate, grade, income, revolving balance, term, utilization of revolving debt, total accounts, and debt-to-income ratio were the strongest drivers of prediction. suggests that loan outcomes in the dataset are shaped by the combined effect of borrowing scale, repayment burden, pricing, existing debt pressure, and overall credit profile rather than by a single variable in isolation. In practice the final model should be treated as a decision-support tool rather than a replacement for human judgment. Its main value is in helping a lender prioritize applications or accounts for closer review, it could support underwriting, pricing decisions, risk monitoring, and portfolio management, however in the end it shouldn't be used as a fully automatic approval rule. In conclusion the supervised part of the project shows us that the future loan performance can be predicted to a meaningful extent using origination-time information, and the strongest result comes from combining a flexible model with a decision threshold chosen for the actual business problem. Together with the unsupervised part of the project, this supervised analysis completes the broader movement from describing borrower groups to predicting realized loan outcomes.